INTELLIGENT ELEPHANT CARE AND VISITOR INSIGHTS: A MULTI-COMPONENT RESEARCH STUDY AT PINNAWALA ELEPHANT ORPHANAGE

Project ID: 25-26J-391

Project Proposal Report

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Sri Lanka Institute of Information Technology Sri Lanka

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ELEPHANT BEHAVIOR & STRESS DETECTION

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DECLARATION PAGE OF THE CANDIDATES & SUPERVISOR

Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Elephants are highly intelligent and socially complex animals, making the ability to understand their emotional states crucial for welfare and conservation. In semi-captive environments such as the Pinnawala Elephant Orphanage in Sri Lanka, traditional observation-based methods are often subjective and lack real-time accuracy. This research addresses this gap by introducing a technology-driven, non-invasive framework for detecting elephant emotions. The proposed system integrates video surveillance, sound analysis, and thermal imaging through Internet of Things (IoT) enabled devices. Using machine learning, the framework will analyze body movements, vocal patterns, and heat variations to identify stress, discomfort, or other emotional states. The methodology prioritizes continuous and real-time monitoring, ensuring minimal disturbance to the elephants while providing reliable insights. Anticipated outcomes include a real-time detection system capable of identifying abnormal emotional responses, such as elevated body heat, unusual vocalizations, or restless movements. These results will help caretakers intervene proactively, enhancing both animal welfare and safety. In conclusion, the study will contribute to objective emotion recognition in elephants, offering practical recommendations for deploying low-cost IoT-based monitoring solutions in conservation settings. This approach has the potential to improve welfare practices not only in Sri Lanka but globally.

Keywords: Elephant Welfare, Emotion Detection, IoT, Multimodal Monitoring

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
CNN	Convolutional Neural Network
R-CNN	Region-based Convolutional Neural Network
CCTV	Closed-Circuit Television
YOLO	Object Detection Architecture

1. INTRODUCTION

Elephants are among the most intelligent and socially complex species in the animal kingdom, exhibiting a wide range of emotions and behaviors that reflect their physical and psychological well-being. Monitoring these emotional states is particularly significant in semi-captive environments such as the Pinnawala Elephant Orphanage in Sri Lanka, where elephants live in close interaction with both caretakers and tourists. Traditional observation-based methods, however, are highly dependent on human judgment, making them subjective, inconsistent, and often unable to capture real-time changes in behavior or physiology. This limitation creates a pressing need for objective, technology-driven approaches to monitor elephant welfare.

Recent advances in Artificial Intelligence (AI), IoT, and sensor technologies offer transformative opportunities to address this gap. Multimodal sensing, which integrates video, acoustic, and thermal data, provides a more holistic representation of animal behavior compared to single-source monitoring systems. By analyzing visual movements, vocalizations, and body heat fluctuations simultaneously, it becomes possible to detect both negative emotional states, such as stress, and positive ones, such as empathy and social bonding.

The research component presented here focuses on developing a multimodal AI powered monitoring system capable of detecting elephant emotions in a non-invasive and real-time manner. Using IoT-enabled devices, data will be collected from Closed-Circuit Television (CCTV) cameras, directional microphones, and thermal sensors placed strategically in the Pinnawala environment. Machine learning algorithms will then be applied to identify behavioral and physiological patterns that correlate with stress or empathetic responses.

This research not only seeks to improve the accuracy of elephant welfare assessments but also to provide a practical, scalable, and energy-efficient solution for conservation environments. By bridging the gap between ethology and technology, the study contributes to the broader goal of integrating advanced AI driven tools into sustainable wildlife management practices.

1.1 Background & literature survey

Elephants are widely recognized for their intelligence, rich social structures, and emotional depth, making the study of their welfare a critical priority in conservation environments such as the Pinnawala Elephant Orphanage in Sri Lanka. Traditional methods of monitoring, primarily reliant on human observation, are often subjective, inconsistent, and incapable of capturing continuous, real-time behavioral or physiological changes. Recent advances in Artificial Intelligence (AI), IoT, and sensor-based technologies have introduced new opportunities for non-invasive welfare monitoring.

For instance, studies of elephant vocalizations demonstrate that rumbles and trumpeting correlate with stress and emotional communication, with AI models successfully detecting vocal patterns even in noisy environments [1]. Similarly, computer vision techniques such as pose estimation and object detection have been applied to video data to identify stress-related movements like ear flapping, head swinging, and pacing, or affiliative gestures like trunk touching, although these methods lack physiological validation [3].

Thermal imaging has also proven effective in detecting stress, as variations in surface temperature particularly in the ears, trunk, and temporal glands serve as indicators of arousal and discomfort [4]. However, each single-modality approach has limitations: acoustic methods suffer from environmental noise, video-based methods may misinterpret context, and thermal data lacks behavioral linkage. Recent research highlights the potential of multimodal approaches, which integrate multiple data streams to provide higher accuracy and robustness [6], yet applications to elephants remain limited.

Most studies continue to emphasize stress detection while neglecting positive emotional states such as empathy, and few systems are designed for practical deployment in resource-constrained environments like Pinnawala. To address this gap, the proposed research develops a multimodal IoT-based AI framework that fuses audio, video, and thermal data to detect both stress and empathetic behaviors in elephants in real time. This novel integration ensures greater reliability, energy efficiency, and practical field deployment, thereby contributing not only to improved animal welfare but also to the advancement of conservation technology.

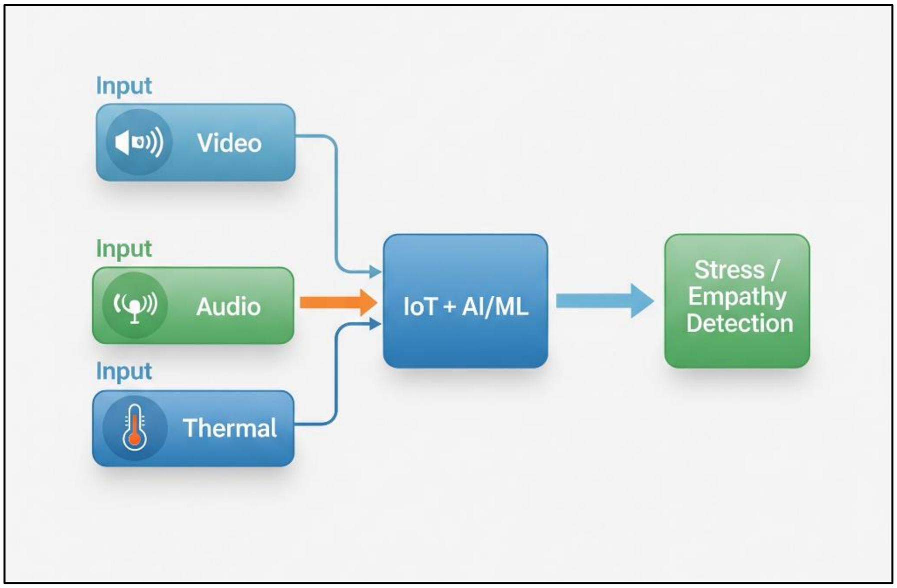

Figure 1.3.1: Framework diagram

Table 1.3.1: Potential Indicators of Stress & Empathy

Modality	Stress Indicators	Empathy Indicators
Audio	Trumpeting, irregular rumbles	Soft rumbles, reassuring calls
Video	Ear flapping, head shaking, pacing	Trunk touching, group protection
Thermal	Elevated ear/trunk temperature (>37.5°C)	Stable/normal heat patterns

Table 2.4.1: Summary of Existing Research Methods

Research	Modality Used	Target Behavior / Stress Indicator	Limitations	Notes
Acoustic features indicate arousal in elephant rumbles (2012) [1]	Audio (Rumbles, Trumpets)	Stress detection, arousal signals	Sensitive to noise, environmental interference	Shows acoustic stress markers
Automated recognition of elephant behavior using computer vision techniques (2021) [3]	Video (Pose estimation, trunk & ear movement)	Stress & social behavior	Misinterpretation without physiology	Strong for visual cues
Infrared thermography in the evaluation of stress and welfare in animals: A review (2022) [4]	Thermal Imaging	Heat changes in ears, trunk	No behavioral context	Non-invasive stress measure
The use of infrared thermography for assessing stress responses in Asian elephants (Elephas maximus) (2022) [5]	Thermal + Hormonal validation	Body surface temperature vs. stress hormones	Limited sample size	Shows link between heat & cortisol

Table 3.5.1: Comparison of Single-Modality vs. Multimodal Approaches

Aspect	Single-Modality Systems	Multimodal (Proposed)
Accuracy	Moderate	High (cross validation)
Noise Resistance	Low (sensitive to environment)	Higher (data fusion)
Behavior + Physiology	Limited (one aspect only)	Combined view
Practical Use in Field	Easier but less reliable	More robust

1.2 Research gap

Table 4.6.1: Research Gap Table

Feature / Method	Acoustic features indicate arousal in elephant rumbles [1]	Automated recognition of elephant behavior using computer vision techniques [2]	The use of infrared thermography for assessing stress responses in Asian elephants (Elephas maximus) [3]	Proposed Multimodal IoT-AI System
Real-time Monitoring	No	No	No	Yes
Audio-based stress detection	Yes	No	No	Yes
Video-based behavioral analysis	No	Yes	No	Yes
Thermal imaging for physiological monitoring	No	No	Yes	Yes
Multimodal Data Fusion (Audio + Video + Thermal)	No	No	No	Yes
IoT-based Deployment	No	No	No	Yes
Detection of stress	Yes	Yes	Yes	Yes

Existing research on elephant stress detection has primarily focused on single-modality approaches, such as analyzing vocalizations (audio-based stress detection), visual behaviors (video-based pose estimation and activity recognition), or physiological cues (thermal imaging of body temperature). While each of these methods has contributed valuable insights, they all suffer from limitations when applied in isolation. For example, audio analysis is highly vulnerable to environmental noise, video monitoring is restricted by lighting and visibility conditions, and thermal imaging alone cannot provide full behavioral context. Moreover, most studies emphasize stress detection only, neglecting the identification of positive emotional states such as empathy, which plays a critical role in elephant social well-being.

To date, no research has combined multimodal data fusion of audio, video, and thermal inputs in a single IoT-powered system for continuous monitoring of elephants in conservation environments. Additionally, practical challenges such as real-time deployment, energy efficiency, and field adaptability have not been fully addressed in existing literature.

The proposed system fills this gap by introducing a multimodal IoT-AI framework capable of fusing behavioral, acoustic, and physiological data to detect both stress and empathy in elephants. This high-level integration not only enhances detection accuracy but also ensures practical applicability in semi-captive environments such as Pinnawala Elephant Orphanage, Sri Lanka. Thus, the novelty lies in moving beyond stress-only detection toward a holistic, real-time, and deployable solution for elephant welfare monitoring.

1.3 Research problem

Elephants are highly intelligent and socially complex animals whose welfare is strongly influenced by their ability to cope with stress. In semi-captive environments such as the Pinnawala Elephant Orphanage, stress monitoring plays a crucial role in ensuring their health, longevity, and quality of life. At present, elephant stress assessment methods depend heavily on human observation, where caretakers interpret visible behaviors such as ear flapping, head shaking, or vocalizations. While these observations are valuable, they are inherently subjective, prone to bias, and cannot provide continuous or real-time data.

In recent years, research has introduced AI-driven systems that rely on single data sources such as audio, video, or thermal imaging. Audio-based systems can recognize vocal stress signals but are highly sensitive to environmental noise. Video-based monitoring captures stress-related body movements but often misinterprets actions when divorced from physiological data. Thermal imaging, though effective in detecting physiological changes such as increased surface temperature, lacks behavioral or contextual information. Each of these modalities, when used alone, provides an incomplete and potentially unreliable picture of elephant stress.

Furthermore, many of these studies remain confined to controlled laboratory conditions or isolated research trials, with little evidence of real-time, field-deployable solutions in conservation environments. There is a significant gap in designing systems that are not only accurate but also scalable, energy-efficient, and adaptable to real-world contexts like Pinnawala, where environmental variability, resource limitations, and continuous monitoring requirements present additional challenges.

Therefore, the research problem is the absence of an integrated, multimodal, IoT-based AI framework that can combine audio, video, and thermal data to deliver continuous, real-time, and objective monitoring of elephant stress. Addressing this problem has the potential to transform welfare monitoring practices by reducing reliance on subjective human observation, enhancing conservation outcomes, and setting a new benchmark for technology-driven animal welfare management.

2. RESEARCH OBJECTIVES

2.1 Main objective

The main objective of this research is to develop and implement a multimodal AI and IOT based framework that integrates audio, video, and thermal data to accurately detect and monitor stress in elephants in real time. The system is designed for practical deployment in semi-captive environments, such as the Pinnawala Elephant Orphanage in Sri Lanka, to support continuous welfare monitoring, reduce reliance on subjective human observation, and enhance evidence-based conservation practices.

2.2 Specific objectives

Collect and preprocess multimodal data from elephants

- This includes gathering audio recordings of vocalizations, video footage of body movements, and thermal imaging of physiological changes from elephants at Pinnawala. The raw data will be cleaned, annotated, and structured into datasets suitable for machine learning model training and evaluation.

Design and train AI models for stress detection

Develop specialized machine learning and deep learning models capable of analyzing different data types:

- Audio models to detect stress-related vocalizations such as trumpeting or rumbling.
- Computer vision models to recognize stress-related body behaviors (e.g., ear flapping, head shaking, pacing).
- Thermal imaging models to identify abnormal heat signatures linked to stress.

Integrate multimodal inputs into a unified AI-IoT framework

- Combine outputs from the individual models (audio, video, thermal) using data fusion techniques to achieve higher detection accuracy. The system will be integrated with IoT devices for real-time data acquisition and transmission, enabling continuous and non-invasive monitoring of elephant stress.

Evaluate the accuracy, efficiency, and reliability of the system

- Conduct experiments to compare multimodal detection results with single-modality approaches, as well as cross-validate system outputs with expert human observations. Key evaluation metrics will include accuracy, sensitivity, specificity, and robustness in field conditions.

Optimize the system for field deployability and conservation use

- Ensure the solution is energy-efficient, cost-effective, and scalable, with the ability to operate in semi-captive environments like Pinnawala. This includes testing the system's durability, minimizing resource requirements, and ensuring it is practical for long-term conservation monitoring.

3. METHODOLOGY

Our research will follow a structured methodology that combines field data collection, AI-based modeling, and system evaluation to develop a multimodal elephant stress detection framework.

Data Collection

The data collection process will focus on gathering multimodal inputs from elephants in semi-captive environments such as Pinnawala Elephant Orphanage. Audio data will be obtained using directional microphones capable of capturing vocalizations such as low-frequency rumbles, trumpets, and distress calls. Simultaneously, video data will be recorded through CCTV cameras to observe behavioral indicators, including ear flapping, trunk movements, head shaking, and pacing, which are commonly associated with stress responses. Thermal data will be collected using IoT enabled thermal imaging sensors to monitor surface body heat, with particular attention to stress-sensitive areas such as the ears and temporal glands. In addition to these modalities, environmental context information such as ambient noise levels, crowd density, and temperature will be documented to support accurate interpretation of the multimodal data and reduce confounding factors.

Data Preprocessing

Once collected, the data will undergo preprocessing to ensure quality and consistency for analysis. Audio signals will be filtered to reduce environmental interference such as human voices, wind, or background noise, after which they will be converted into spectrograms for model input. Video data will be segmented into frames and pose estimation techniques will be applied to identify and track key behaviors across sequences. Thermal data will be normalized to account for environmental temperature variations, enabling the detection of deviations from the normal elephant body temperature range of 35–36°C. All three modalities will then be synchronized to create a unified dataset, allowing cross validation of stress indicators across different data types.

Model Development

The processed data will be used to train specialized AI models for each modality. Audio-based stress detection will employ Convolutional Neural Network (CNN) trained on spectrogram features to classify different types of vocalizations. Video data will be analyzed using advanced computer vision models, such as YOLO or Mask R-CNN, to detect elephants, track movement, and identify abnormal stress-related behaviors. For thermal data, classification models will be designed to detect elevated or irregular heat signatures in stress sensitive regions. These independent models will form the basis for a multimodal stress detection framework.

Multimodal Integration

To improve reliability and accuracy, the outputs from the audio, video, and thermal models will be integrated into a multimodal fusion framework. Ensemble learning techniques will be applied to assign weights to different modalities depending on context, ensuring that one source of data can compensate for the limitations of another. For example, when external noise reduces audio reliability, greater weight may be assigned to video and thermal data. This fusion approach is expected to provide a more holistic and accurate assessment of elephant stress compared to single-modality systems.

System Evaluation

The performance of the proposed system will be evaluated using both quantitative and qualitative methods. Metrics such as accuracy, precision, recall, and F1-score will be calculated to measure the effectiveness of the AI models. Comparative analysis will be conducted between single-modality models and the proposed multimodal framework to demonstrate improvements in detection accuracy. Additionally, field validation will be carried out at Pinnawala Elephant Orphanage, where system outputs will be compared against expert observations by elephant caretakers and veterinarians.

Deployment

Following evaluation, the system will be deployed in selected areas of Pinnawala as a real-time monitoring framework. IoT-enabled devices, including microphones, cameras, and thermal sensors, will continuously capture data and feed it into the AI models through an IoT gateway. The results will be displayed on a centralized dashboard, providing real-time alerts, behavioral trends, and stress assessments for conservation staff. This practical deployment ensures that the research not only contributes to academic knowledge but also offers a tangible conservation tool for enhancing elephant welfare.

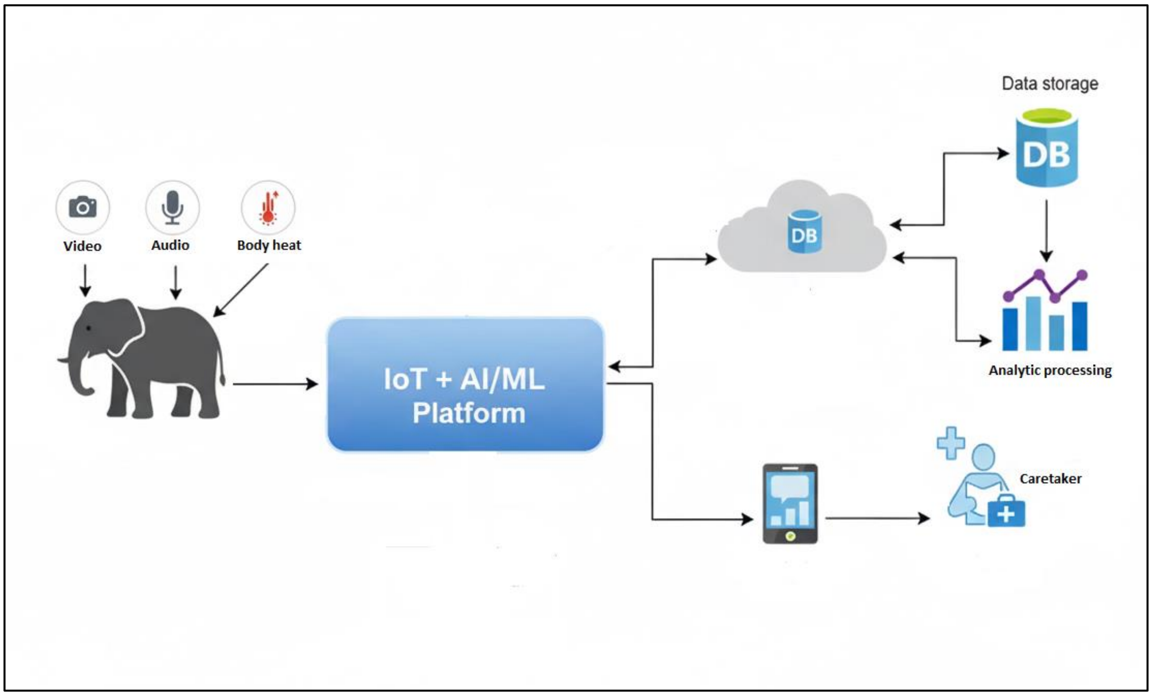

Figure 2.13.1: System Overview diagram

3.1 Testing and validations

The system will be tested in two stages, first using pre-recorded datasets of elephant vocalizations, behavioral videos, and thermal images to validate model accuracy under controlled conditions, and then through live deployment at Pinnawala Elephant Orphanage to assess performance in real environments. Predictions from the system will be compared with expert observations by caretakers and veterinarians to ensure reliability. Key evaluation metrics such as accuracy, precision, recall, and F1-score will be calculated, while IoT devices will also be tested for energy efficiency, robustness, and real-time monitoring capabilities

3.2 Project requirements

3.2.1 Functional Requirements

- The system must capture video and audio data through CCTV cameras and microphone to track behavioral indicators (ear flapping, trunk movement, pacing) and to monitor vocalizations (rumbles, trumpeting, distress calls).
- The system must capture thermal data via IoT-enabled thermal sensors to monitor body heat changes.
- The system must process and fuse audio, video, and thermal data using AI/ML algorithms.
- The system must detect and classify stress-related behaviors in elephants.
- The system must generate alerts for abnormal stress conditions.
- The system must store data securely for further analysis.

3.2.2 Non-Functional Requirements

- **Accuracy**: Stress detection must achieve at least 85% precision and recall.
- **Scalability**: The system should handle multiple elephants simultaneously.
- **Reliability**: IoT devices must function in outdoor environments with minimal downtime.
- **Energy Efficiency**: Sensors and devices must be optimized for long-term deployment.
- **Security**: Collected data must be stored and transmitted securely.
- **Usability**: The system interface must be simple for caretakers and veterinarians to use.

3.2.3 User Requirements

- Caretakers should be able to view stress alerts in real time.
- Veterinarians should be able to access historical data for health analysis.
- The system should require minimal technical knowledge for daily use.

3.3 High-Level System Architecture & Self-Evaluation Plan

The proposed system adopts a four-layered architecture,

(1) Data Collection - CCTV with integrated microphones for synchronized audio, video, motion and IoT thermal sensors for body-heat monitoring.

(2) Preprocessing and Signal Processing - to clean and prepare multimodal inputs. (noise reduction, spectrogram generation, frame extraction, motion heatmaps, thermal normalization).

(3) AI/Decision Layer – modality specific deep models audio CNNs, YOLO/CNN for video & motion, thermal anomaly detector and a multimodal fusion engine.

(4) Output & Interface - Flutter dashboard for real-time alerts, logging.

The self-evaluation plans the system will be tested using several clear measures. First, its accuracy will be checked by comparing results to real labeled data, using metrics like precision, recall, and F1-score. Next, speed and reliability will be measured, making sure alerts are given within seconds and false alarms stay low, even in noisy or crowded environments. The system's robustness will be tested by running it on data from different elephants, days, and settings to confirm consistency. We will also evaluate usability by getting feedback from caretakers and veterinarians on how useful and timely the alerts are in practice. Finally, ethical and safety checks will ensure the system is non-invasive, does not harm the elephants, and protects collected data. Overall, these evaluations will prove that the system is accurate, fast, reliable, user-friendly and suitable for real-world deployment in conservation.

4. DESCRIPTION OF PERSONAL AND FACILITIES

The research will be conducted with the combined expertise of the research team and the support of facilities available at both the university and the Pinnawala Elephant Orphanage. The research team consists of undergraduate research members specializing in Information Technology, with knowledge in AI, Machine Learning (ML), IOT, and data analytics. The team members are responsible for system design, data collection, model training, integration, and evaluation. Supervision and guidance will be provided by academic staff with expertise in computer vision, bioacoustics, and conservation technology, ensuring both technical rigor and scientific validity.

Programming language

- **Dart (Flutter) -** For frontend application. Provides real-time visualization of stress indicators, graphs, and alerts.
- **Python** - For AI/ML model development**.**
- **C / C++ -** Used in IoT microcontrollers (ESP32 / Arduino) to read thermal sensors and transmit data via MQTT/Node-RED.
- **Firebase** – For the database.

Hardware Components

- **Directional Microphones** – to capture elephant vocalizations (rumbles, trumpets, distress calls).
- **CCTV** – for recording elephant behavior such as ear flapping, pacing, and trunk movements.
- **Thermal Sensors** – to monitor body heat patterns, especially ears and temporal glands.
- **IoT Devices** – for sensor integration, data transmission, and energy-efficient edge processing.

System Technologies

- **AI & ML -** CNN, Mask Region-based Convolutional Neural Network (R-CNN), and audio classification models for stress detection.
- **Computer Vision -** Object detection and pose estimation to identify stress-related behaviors from video streams.
- **IoT Integration -** Real-time data transmission from microphones, cameras, and thermal sensors to central servers/cloud.
- **Data Fusion Techniques -** Multimodal integration of audio, video, and thermal signals for accurate stress detection.
- **Evaluation Tools -** Statistical analysis, accuracy testing, and visualization dashboards to monitor system outputs.

Commercialization

The proposed system demonstrates strong business potential by addressing the growing demand for non-invasive, technology-driven animal welfare solutions. Its immediate application lies in elephant conservation centers, zoos, and orphanages, where real-time monitoring reduces staff workload and ensures timely interventions. Beyond elephants, the system can be scaled to other wildlife species, broadening its market scope.

From a commercialization perspective, the system is cost-effective, integrates seamlessly with existing infrastructure (CCTV, IoT) and is ethically compliant making it attractive for both governments funded conservation projects and private organizations.

User benefits include**:**

- Real-time stress detection and early intervention.
- Reduced manual observation workload.
- Objective, data-driven decision-making.
- Improved animal welfare and conservation outcomes.
- Scalability to other species and global markets.

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APPENDICES

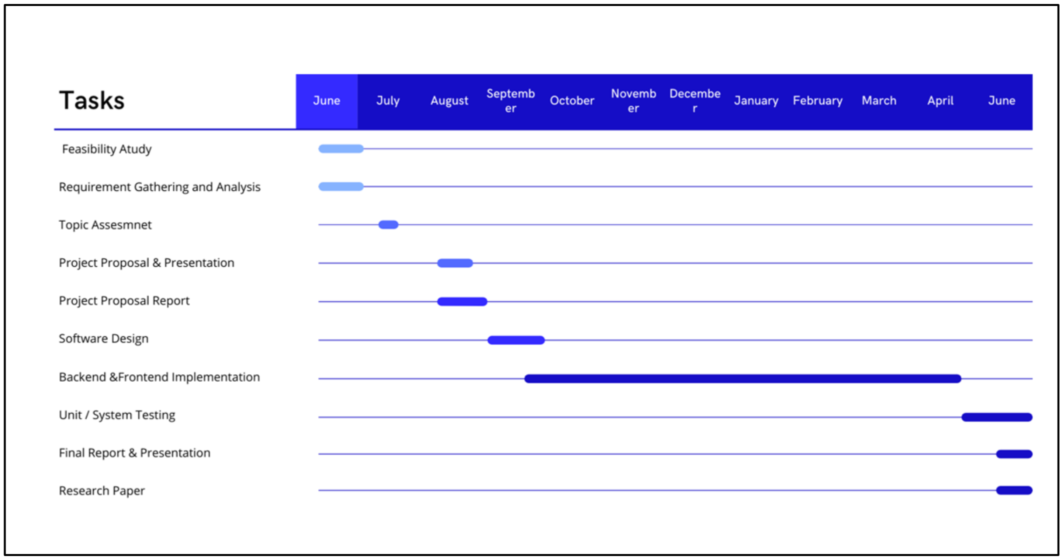

Figure 3.19.1: Gantt Chart

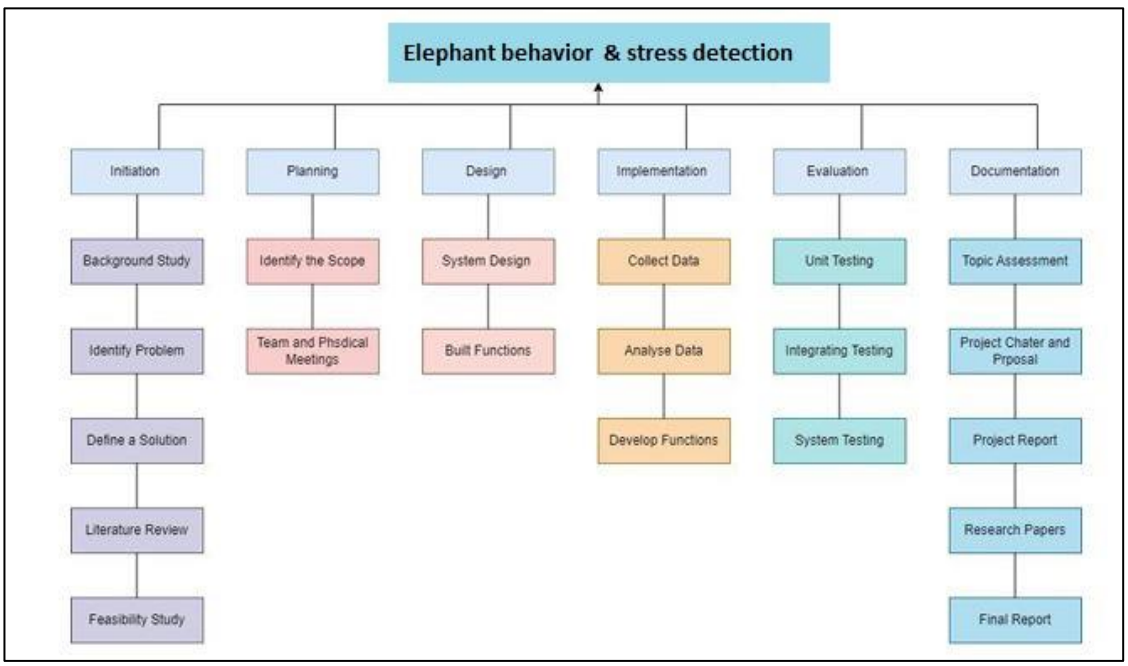

Figure 4.19.2: Work breakdown structure

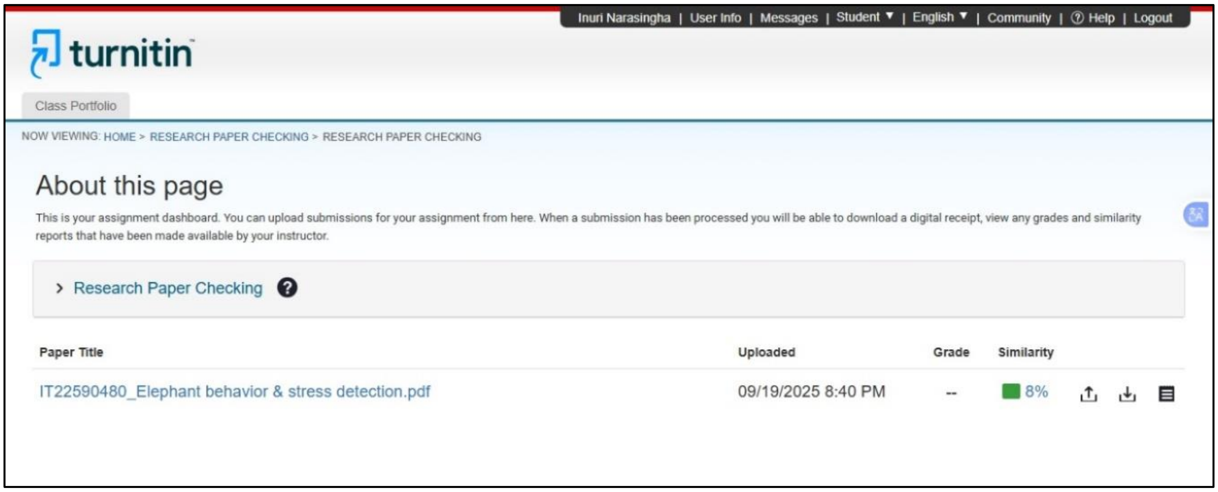

Figure 5.20.1: Plagiarism Report